In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from PIL import Image
```

In [3]:
```
image = Image.open('image.jpg')

print(image.size)
```

```
(735, 412)
```

In [4]:
```
image = image.resize((396, 396))

print(image.size)
```

```
(396, 396)
```

In [5]:
```
img = np.array(image)

print(img.shape)
```

```
(396, 396, 3)
```

In [6]:
```
pixels = img.reshape(-1, 3)

print(pixels.shape)
```

```
(156816, 3)
```

In [7]:
```
model = KMeans(
    n_clusters=16,
    init='k-means++',
    random_state=42,
    n_init=10
)

model.fit(pixels)
```

Out[7]:

▼ KMeans (i) (?)

▶ Parameters

▶ Fitted attributes

▸ 16 features

In [8]:
```
labels = model.predict(pixels)

print(labels[:20])
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0]
```

In [9]:
```
colors = model.cluster_centers_

print(colors)
```

```
[[220.85248423 215.03544953 190.41088328]
 [137.9055528  157.69955379 159.50322261]
 [130.04694836  75.16431925  76.81220657]
 [250.65370216 241.91763727 202.89330283]
 [ 83.41460317 104.03047619 125.58031746]
 [187.37720553 178.03051979 167.35288507]
 [164.3431085   97.92492669  94.70791789]
 [207.15435606 135.30681818 116.02651515]
 [166.59134615 149.99198718 140.70833333]
 [169.92706175 191.12805636 181.37629507]
 [151.327516   174.55642816 171.55119255]
 [118.86451613 122.48064516 129.14301075]
 [241.25735568 229.59180633 197.5273743 ]
 [ 95.74541752 139.18533605 156.16089613]
 [194.19273892 205.40576615 187.36946076]
 [128.69961427 142.74011572 146.63982642]]
```

In [10]:
```
compressed_pixels = colors[labels]

compressed_pixels = compressed_pixels.astype(np.uint8)
```

In [11]:
```
compressed_image = compressed_pixels.reshape(396, 396, 3)
```

In [12]:
```
plt.figure(figsize=(10, 5))

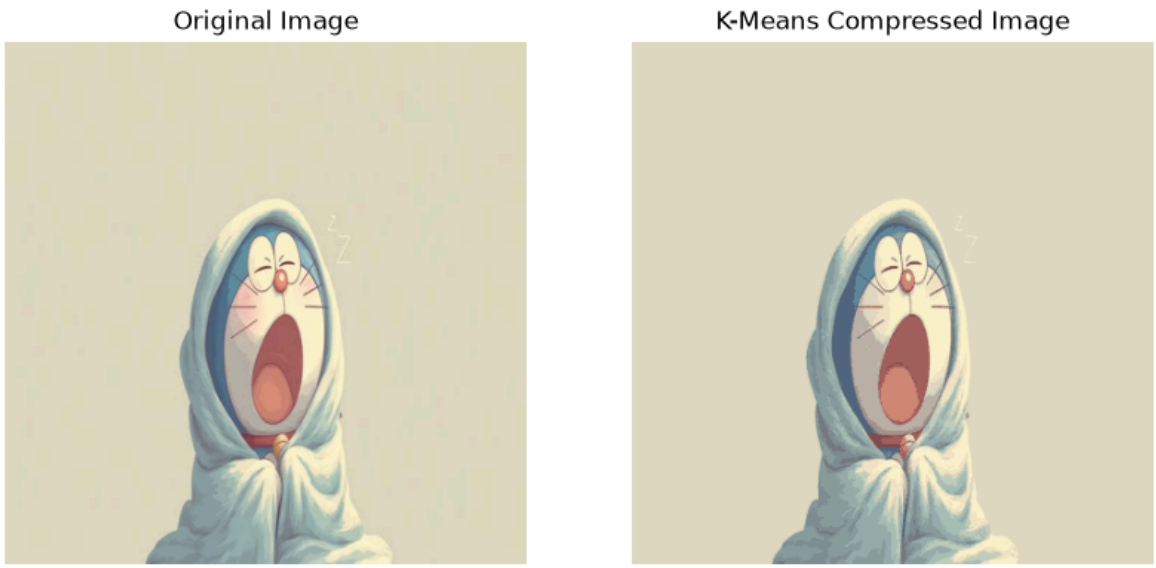
plt.subplot(1, 2, 1)

plt.imshow(img)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 2, 2)

plt.imshow(compressed_image)
plt.title('K-Means Compressed Image')
plt.axis('off')

plt.show()
```




In [ ]:
```

```